Windows Network Troubleshooting & Diagnostics

Practical Technical Support Lab | Windows PowerShell | 24 October 2026

1. Project overview

Objective: Verify local network configuration, test gateway and internet connectivity, investigate DNS name resolution, and trace the route to a public destination. Findings are based on actual PowerShell outputs supplied during this lab.

Environment: Windows laptop; MediaTek Wi-Fi 6E MT7902 wireless adapter; PowerShell and built-in Windows networking utilities.

2. IP configuration findings

Field	Observed value
IPv4 address	192.168.1.95 (private LAN address)
Subnet mask	255.255.255.0
Default IPv4 gateway	192.168.1.1
DHCP	Enabled
IPv4 DNS servers	103.21.187.19; 103.21.187.18
Default DNS queried by nslookup	fe80::1 (IPv6 link-local address)

Interpretation: The active Wi-Fi adapter had a private IPv4 address, a gateway, and configured DNS resolvers. This alone does not establish external connectivity.

3. Connectivity testing - ping

Destination	Packets received	Loss	Average RTT	Finding
192.168.1.1 (gateway)	4 of 4	0%	2 ms	Reachable
8.8.8.8 (public IP)	4 of 4	0%	5 ms	Reachable
google.com (IPv6)	4 of 4	0%	7 ms	Reachable

Interpretation: All three tested destinations responded to ICMP echo requests. These are four-packet samples, not long-term reliability measurements. The domain-based ping also shows that name resolution succeeded through at least one Windows resolution path.

4. DNS resolution investigation

Test	Observed result
nslookup google.com (default resolver)	DNS request timed out; server displayed as UnKnown at fe80::1.
nslookup google.com 8.8.8.8	Successful response from dns.google (8.8.8.8); returned IPv4 and IPv6 addresses.

Assessment: The default resolver did not answer this nslookup attempt, whereas the explicitly selected Google Public DNS server did. This suggests a resolver-specific issue or transient timeout, but does not establish the root cause. The word UnKnown can also appear when a DNS server name is unavailable; the timeout is the relevant failure.

Recommended follow-up: Repeat both lookups, test configured IPv4 DNS servers directly, and compare results before making DNS configuration changes. Do not change company-managed network settings without authorization.

5. Route tracing

Command: tracert google.com

Observed: IPv6 destination 2404:6800:4013:813::8a reached at hop 21. Intermediate hops 12 through 20 showed request timeouts, but the final destination responded and the trace completed.

Interpretation: Intermediate devices can suppress traceroute replies; timeouts along the path do not necessarily indicate an outage when the destination is reached.

6. Commands used

ipconfig /all
ping 192.168.1.1
ping 8.8.8.8
ping google.com
nslookup google.com
nslookup google.com 8.8.8.8
tracert google.com

7. Troubleshooting summary

Check	Outcome
LAN gateway connectivity	Passed
External IP connectivity	Passed
Domain ping	Passed
Default DNS lookup	Timed out during test; requires follow-up
Google Public DNS lookup	Passed
Traceroute to destination	Completed

8. Conclusion

The laptop had working connectivity to its gateway and tested internet destinations at the time of the checks. A default-DNS nslookup timed out, but querying Google Public DNS directly succeeded. This lab demonstrates a practical support workflow: identify configuration, isolate connectivity versus name-resolution behavior, compare DNS resolvers, and document evidence without overstating the diagnosis.

Portfolio-ready description: Conducted a Windows network diagnostics lab using PowerShell, ipconfig, ping, nslookup, and tracert; documented connectivity metrics and investigated a default DNS resolver timeout through comparative DNS testing.

Privacy note: This report intentionally omits MAC addresses, host identifiers, and full device-specific IPv6 addresses. Review raw command outputs and screenshots before publishing them publicly.